

Wallwright and Useful

A head to head, and a correction to an earlier recommendation.

Prepared for the Wallwright exhibit · 24 August 2026 · **VERIFIED** marks a claim a cited source states. **UNVERIFIED** marks inference or something not checked. Useful's architecture is quoted from Useful's own documentation.

A correction

An earlier comparison ranked Useful the best commercial fit for this exhibit, largely on published pricing and documented tablet control. Reading Useful's own hardware documentation changes that. Useful is **not software you install on the show PC**. It is a Linux operating system, on its own certified server, streaming video to a decoder device behind each display. For a single wall driven by a single Windows PC, it is the wrong shape, and its rendering model is the one this project rejected at the design stage.

What each one actually is

Useful

A platform, not an application. From Useful's documentation:

"Useful on-premise is, at its core, a complete Linux-based operating system derived from Red Hat Enterprise Linux (RHEL) 9." **VERIFIED**

Content reaches displays indirectly. Sources are ingested at the server, in part through Blackmagic DeckLink capture cards for HDMI and SDI inputs, and then **streamed to uClient endpoint devices over RTSP**. A uClient runs on a smart display's own processor or on an Android adapter behind each screen. **VERIFIED**

The server needs a certified NVIDIA Quadro or RTX A-series GPU, from a list that runs RTX 4000 through A6000, because the GPU decodes and encodes the video. Deployments past twelve displays need 10 Gb network interfaces. **VERIFIED**

Wallwright

An application. It runs on the show PC, on Windows, and draws each panel as a real browser view composited directly onto the wall's output. Nothing is encoded, streamed or decoded. Input reaches a panel the way it reaches a browser tab, so clicking and typing need no remote-control layer. **VERIFIED**

What deploying each one looks like

	USEFUL	WALLWRIGHT
Machine	A dedicated server running Useful's own RHEL 9 derived OS. It cannot share the show PC with Windows. VERIFIED	The existing Windows show PC. VERIFIED
GPU	A certified NVIDIA Quadro or RTX A-series card. VERIFIED	Whatever drives the wall already.
Behind the display	A uClient device, or a smart display running the uClient app. VERIFIED	Nothing. The PC's own output.
Network	Endpoints on the LAN, 10 Gb interfaces past twelve displays. VERIFIED	None required. Optional LAN for the tablet control page.
How a page reaches the wall	Rendered at the server, encoded, streamed over RTSP, decoded at the endpoint. VERIFIED	Rendered on the wall's own GPU. No encode or decode. VERIFIED
Licence	\$500 per display, per year. VERIFIED	None.

Why the streaming matters here, specifically

SPEC.md considered capture based approaches at the outset and rejected them, because what lands on the wall is a video texture and native interaction is lost: making it interactive means mapping wall coordinates back into an offscreen browser and injecting synthetic input. Useful's architecture is a well engineered version of exactly that shape, with the interaction added back through its own remote control layer. That is a reasonable trade for a control room streaming cameras and desktops across a building. It is a strange trade for one wall showing four web pages, sitting on top of the PC that could simply render them.

Where Useful clearly wins

- **Anything that is not a web page.** Cameras, HDMI, SDI, remote desktops, video conferencing. Wallwright does web pages and nothing else. VERIFIED
- **More than one wall, or more than one room.** This is what the architecture is for, and it is why the architecture looks like that.
- **Redundancy.** A second failover server. Wallwright has none, and on one PC cannot have one. VERIFIED
- **A support contract and a maintained product** that outlives whoever wrote the exhibit.
- **Mature remote control**, including operators driving mouse and keyboard from a desktop or smartphone. VERIFIED

Where Wallwright wins for this exhibit

- **Native rendering and native input.** No encode, decode, stream or coordinate mapping between the operator's hand and the page.
- **It runs on the machine that is already there.** No second server, no endpoint device, no operating system change, no GPU purchase.
- **Arranging the montage on the live wall.** Drag panels around against the real dashboards, at the real resolution. Useful's layouts are authored in a console. **UNVERIFIED THAT USERFUL CANNOT DO THIS**
- **Several panels behind one login.** Four dashboards from one SSO-protected application share a session rather than needing four sign-ins. No vendor documents an equivalent. **UNVERIFIED**
- **No recurring cost and no vendor dependency.**

The honest comparison

	USERFUL	WALLWRIGHT
Live interactive web pages	Yes, via streaming plus remote control	Yes, natively
Runs on the existing show PC	No	Yes
Extra hardware needed	Server, GPU, endpoint per display	None
Handles an SSO login on the wall	Not documented	Yes, measured
Panels sharing one login	Not documented	Yes
Edit the montage at the wall	Console authored	Yes
Named layout presets	Yes	Yes
Tablet and phone control	Yes, mature	A web page served by the wall
Non-web sources	Yes	No
Multi-wall, multi-room	Yes	No
Failover	Yes	No
Vendor support	Yes	No
Recurring cost	\$500 per display per year	None
Proven at this exact job	Needs a demo	Proven except on Windows compositing

Recommendation

For this exhibit as scoped, Wallwright is the better fit, and the reason is architectural rather than financial. Useful would mean replacing the show PC with a Linux server, buying a certified GPU and an endpoint device, and then streaming a web page to a display that is already attached to a computer capable of rendering it. The \$500 licence is the smallest part of that.

Useful becomes the right answer if any of these turn out to be true:

1. The exhibit needs camera, HDMI or desktop sources, not only web pages.
2. It grows to several walls or rooms.
3. Unattended reliability with failover becomes a hard requirement.
4. Honeywell will not accept an unsigned in-house application on the show PC, but will accept a commercial platform. That is a real possibility and worth asking early.
5. **Overlay compositing fails on Windows.** This is still the one unverified assumption underneath Wallwright, and the fallbacks are more awkward. If it fails and the fallbacks disappoint, the build versus buy question genuinely reopens.

What is still unverified

- Whether Useful can drive displays directly from the server's own outputs rather than through streamed endpoints. Zero clients were deprecated in version 12.7, and the current documentation describes streaming. A direct-attached configuration may exist. **UNVERIFIED**
- Whether Useful handles a real Honeywell SSO flow. No vendor documents this.
- How interaction feels through a stream. Latency was not measured and may be perfectly acceptable.
- Wallwright's overlay compositing on Windows, which remains the largest open risk in the project.

Sources

1. Useful, self-sourcing a server: OS, GPU, network, RTSP streaming to uClients. <https://docs.userful.com/userful-infinity/hardware-overview/supported-servers/self-sourcing-a-userful-server.md>
2. Useful, video wall software. <https://www.userful.com/video-wall>
3. Useful, beyond video wall processors. <https://www.userful.com/video-wall-processor>
4. Useful, zero client setup and 12.7 deprecation. <https://docs.userful.com/support-portal/hardware-setup/client-setup/zero-client-setup>
5. Useful pricing, SHI listing. <https://www.shi.com/product/32215340/userful-Professional-Video-Wall-Deployment-Solution>
6. Wallwright design rationale: SPEC.md in the repository.
7. Wallwright measurements: docs/validation.md in the repository.